

Title	<i>Developing a domain-specific LLM through self-instruct fine-tuning</i>
Contacts of the person(s) proposing the project	Zhiguo Zeng: zhiguo.zeng@centralesupelec.fr (Please have a meeting with the supervisor before your select this project!)
Institution (Company, lab,...)	LGI, Centralesupelec

Project description

• Context (1/2 page):

Engineers are interested in reliability, risk, and safety of complex systems like nuclear powerplant, airplane and electrical grid, in order to ensure the correct functioning these essential systems. Assessing reliability, risk, or safety requires modeling systems behavior and leveraging mathematical tools for computing meaningful indicators and simulating scenarios. Currently, this requires a lot of programming effort mainly conducted manually by experienced reliability engineers.

Large Language Models (LLMs) have convincingly proved that they can handle general programming tasks quite well (for example, see Github Copilot and Tongyi Lingma from Alibaba). In previous years, our students have attempted to leverage state-of-the-art LLMs to assist with reliability-related coding tasks. A dataset of around 50 reliability coding tasks was created with unit tests to evaluate the performance of the LLM. However, the results from state-of-the-art LLMs were consistently poor. The models frequently provided incorrect or nonsensical code, failed to grasp the nuances of reliability concepts, and could not reason through multi-step problems in this domain. This performance gap is a direct result of the LLMs' lack of exposure to specialized reliability engineering knowledge during their general pre-training. This project is a direct response to this limitation, aiming to bridge the knowledge gap by developing a fine-tuned LLM specifically for the reliability domain.

• Goals (1/2 page):

The primary goal of this project is to develop a highly effective, domain-specific large language model for reliability engineering. We will achieve this by creating a large-scale, high-quality, synthetic dataset and using it to fine-tune a pre-trained LLM. This approach addresses the core problem of knowledge scarcity and promises to significantly improve the model's performance on complex, domain-specific coding questions. The project will be structured around the following key goals:

1. **Synthetic Data Generation:** We will use the **self-instruct fine-tuning** strategy to automatically generate a large corpus of question-and-answer pairs related to reliability engineering^{1,2,3}. This will be achieved by prompting a powerful LLM to act as a data generator. It will ingest text from our collection of millions of reliability textbook tokens and generate a diverse range of coding questions and corresponding solutions.
2. **Chain-of-Thought (CoT) Integration:** For each generated coding problem, we will automatically generate a **Chain-of-Thought** reasoning process⁴. This multi-step reasoning will break down the solution into a logical sequence, explaining the intermediate steps. This approach is critical for teaching the model how to reason, not just how to provide an answer, which is essential for complex reliability problems.

3. **Model Fine-Tuning and Evaluation:** We will use the resulting high-quality dataset to fine-tune a pre-trained LLM. The fine-tuned model's performance will be rigorously evaluated against our initial hand-written dataset of 50 reliability coding questions, which served as a baseline. The project's success will be measured by the significant improvement of the fine-tuned model on this benchmark compared to its out-of-the-box performance.

- **References:**

1. Ranaldi, L. & Freitas, A. *Self-Refine Instruction-Tuning for Aligning Reasoning in Language Models*. in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing* 2325–2347 (2024). doi:10.18653/v1/2024.emnlp-main.139.

2. Ding, B. et al. *Data Augmentation using Large Language Models: Data Perspectives, Learning Paradigms and Challenges*. Preprint at <https://doi.org/10.48550/arXiv.2403.02990> (2024).

3. Wang, Y. et al. *Self-Instruct: Aligning Language Models with Self-Generated Instructions*. Preprint at <https://doi.org/10.48550/arXiv.2212.10560> (2023).

4. Zhang, Z., Zhang, A., Li, M. & Smola, A. *Automatic Chain of Thought Prompting in Large Language Models*. Preprint at <https://doi.org/10.48550/arXiv.2210.03493> (2022).

- **Expected work and deliverables**

1. Final report
2. Publication if possible
3. Fine-tuned LLMs for vertical domain of risk, reliability and resilience.
4. A github repository of your work.

- **Technical aspects**

You will be provided an evaluation baseline, developed from previous projects. The evaluation baseline comprises of ~50 hand-written coding questions related to reliability engineering. Your goal is to first to reproduce the work of previous year students: Run the LLM to generate code, and evaluate their performance. Then, you will be presented a dataset of reliability textbooks (already pre-processed). Your goal is to implement self-instruct strategy to synthesis data to fine-tune the LLM to perform better in the domain-specific coding tasks.

Comments (e.g. data access, specifics...)

- **Resources made available**

You will be given some sample questions for developing the evaluation baseline. You will work with one PhD student on collecting the data for and the implementation of fine-tuning. Also, a Github repository from previous year will give you a first version of the evaluation baseline, and also how to use it to evaluate the performance of current LLMs.